

# signal-processing.md

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## Signal Processing — The Science Behind BioSync

### Why HRV Variance, Not Heart Rate

Most rPPG research targets **heart rate (HR)** extraction — counting beats per minute.

BioSync targets something different: **short-term Heart Rate Variability (HRV) proxy** — the variance in the beat-to-beat timing.

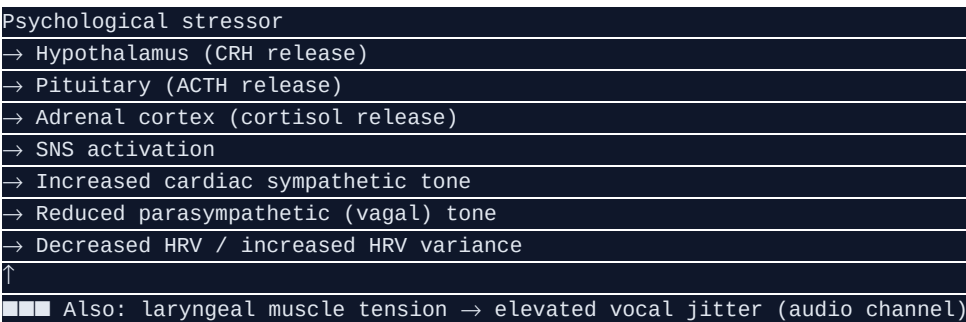
This distinction is critical and often missed:

| Metric           | What it measures             | Stress detection speed               | Lighting sensitivity          |
|------------------|------------------------------|--------------------------------------|-------------------------------|
| Heart rate (BPM) | Mean cardiac cycle frequency | Slow (~30-60s for reliable estimate) | High (absolute values vary)   |
| HRV variance     | Cycle-to-cycle irregularity  | Fast (~1-3s window)                  | Low (delta-based, normalized) |

Under sympathetic nervous system (SNS) activation — the physiological stress response — HRV *decreases* and becomes more erratic. The signal becomes both noisier and less periodic.

This is detectable in ~1 second of signal, whereas reliable HR extraction requires ≥5 seconds.

The HPA axis pathway:



Both the visual (rPPG) and audio (jitter) channels measure downstream effects of the same upstream cortisol event — via independent physiological pathways.

This orthogonality is why multimodal fusion is more robust than either channel alone.

### Remote Photoplethysmography (rPPG)

#### The Physical Principle

Oxygenated hemoglobin (HbO<sub>2</sub>) has characteristic absorption peaks across the visible spectrum.

At ~530–550nm (green light), absorption is significantly higher than for deoxygenated hemoglobin:



|                    |           |                                             |
|--------------------|-----------|---------------------------------------------|
| ~420nm (violet)    | Very high | Strong but noisy, motion-sensitive          |
| ~530-550nm (green) | High      | ■ Optimal SNR for consumer cameras          |
| ~660nm (red)       | Lower     | Used in pulse oximeters (transmission mode) |
| ~940nm (infrared)  | Very low  | Used in smartwatch PPG                      |

With each heartbeat, subdermal blood volume increases, temporarily increasing HbO<sub>2</sub> concentration in capillaries. The camera captures slightly less reflected green light during systole (peak blood flow) than diastole. This variation is ~0.5–2% of total green channel intensity — invisible to the human eye, measurable across a large ROI.

## References:

- Verkrusse, W., Svaasand, L.O., Nelson, J.S. (2008). Remote plethysmographic imaging using ambient light. *\*Optics Express\**, 16(26), 21434–21445.
- de Haan, G., & Jeanne, V. (2013). Robust pulse rate from chrominance-based rPPG. *\*IEEE Transactions on Biomedical Engineering\**, 60(10), 2878–2886.

## Implementation Pipeline

### Step 1: Face Detection and ROI Selection

**File:** signal-processing/roiDetector.ts

MediaPipe Face Mesh provides 468 normalized landmarks per frame, running as a WASM module with no GPU requirement and no network calls after initial load (~16ms/frame).

```
// Forehead landmark subset – chosen for highest SNR and lowest motion artifact
export const PRIMARY_FOREHEAD_LANDMARKS = [10, 151, 9, 8, 107, 336];
// 10 = top-center forehead
// 151 = left-center forehead
// 9 = near hairline, center
// 8 = between brows, upper
// 107 = left inner brow
// 336 = right inner brow
export function extractForeheadROI(landmarks, frameWidth, frameHeight): ROIQuality {
  // AABB with 5% padding around landmark set
  // Falls back to extended set [+54, 284, 103, 332] if area < 400px²
  // Returns: { roi, coverageRatio, sufficient }
}
export function isHeadPoseAcceptable(landmarks): boolean {
  // Yaw proxy: z-depth differential between L/R eye corners
  // Rejects frames where yaw > ~25° (ROI exits frame, SNR drops)
  const yawProxy = Math.abs(landmarks[33].z - landmarks[263].z);
  return yawProxy < 0.15;
}
```

### Why the forehead?

- Largest continuous flat skin surface → largest ROI → most pixels → best SNR
- Perfused by supratrochlear and supraorbital arteries — strong pulsatile signal
- Zero jaw/cheek muscle activity during typing (unlike cheek ROI)

- Minimal occlusion (glasses sit below, hair above)
- Consistent landmark geometry across ethnicities and face shapes

## Step 2: Green Channel Extraction

**File:** signal-processing/rppgExtractor.ts, `hooks/useRPPG.ts`

```
// RGBA pixel layout: [R0, G0, B0, A0, R1, G1, B1, A1, ...]
// Green channel = every 4th element starting at index 1
function extractGreenMean(data: Uint8ClampedArray): number {
  let sum = 0;
  for (let i = 1; i < data.length; i += 4) sum += data[i];
  return sum / (data.length / 4); // Mean green intensity: 0-255
}
```

One sample per video frame → ~30 samples/second → 1800 samples/minute.

The cardiac frequency band is 0.75–2.5 Hz (45–150 BPM) — well within the 15 Hz

Nyquist limit of a 30fps signal.

## Step 3: Signal Smoothing (Two-Pass)

**File:** utils/kalmanFilter.ts

### Pass 1 — 1D Kalman Filter:

The Kalman filter is the optimal recursive estimator for linear systems with Gaussian noise.

For rPPG, it suppresses webcam compression artifacts and micro-movement noise while preserving the genuine pulsatile component.

```
Prediction:  $P(t) = P(t-1) + Q$ 
Update gain:  $K = P(t) / (P(t) + R)$ 
State update:  $x(t) = x(t-1) + K \times (z(t) - x(t-1))$ 
Error update:  $P(t) = (1 - K) \times P(t)$ 
```

Parameters:

- `Q = 0.1` (process noise) — the true signal changes slowly; low process noise is appropriate
- `R = 1.0` (measurement noise) — individual frames are noisy; high measurement noise → aggressive smoothing

The resulting Kalman gain  $K \approx 0.09$  at steady state — each new measurement moves the estimate ~9% of the residual. This provides excellent noise rejection while maintaining responsiveness to genuine physiological changes (~1–2s lag).

For high-motion demo environments (nervous presenter), increase R to 2.0–3.0.

### Pass 2 — 5-sample Moving Average:

Applied after Kalman to remove residual quantization noise. Window = 5 samples = 167ms at 30fps.

Computationally trivial ( $O(n)$ ,  $n=5$ ). Combined effect: the signal is clean enough for reliable variance computation.

#### Step 4: Variance-Based Stress Feature

**File:** `utils/kalmanFilter.ts` — ``computeNormalizedVariance()``

```
//  $V = \sigma^2(G[t-window:t]) / \mu(G_{baseline}) \times 1000$ , clamped 0-100
function computeNormalizedVariance(buffer, baseline, window = 30): number {
  const slice = buffer.slice(-window); // Last 1s of signal
  const variance = slice.reduce(
    (acc, v) => acc + Math.pow(v - baseline, 2), 0
  ) / slice.length;
  // Coefficient of variation  $\times 1000 \rightarrow$  maps typical rPPG variation to 0-100
  return Math.min(100, (Math.sqrt(variance) / baseline) * 1000);
}
```

#### Why this is lighting-independent and skin-tone-independent:

Raw green channel values vary by  $\pm 30\%$  between users (skin tone) and  $\pm 20\%$  per session (lighting). Dividing by the personal baseline (established during 3-second calibration) normalizes for both. The metric becomes: *\*how much is your signal deviating from your own normal?\** — not from any universal constant.

This removes two major sources of demographic bias present in naive rPPG implementations.

#### Step 5: Signal Confidence Scoring

**File:** `hooks/userRPPG.ts`, ``utils/stressEngine.ts``

```
// SNR proxy from short-window coefficient of variation
const recent = buffer.slice(-10); // ~333ms at 30fps
const recentMean = mean(recent);
const recentStd = stddev(recent, recentMean);
const snr = recentMean / (recentStd +  $\epsilon$ ); // higher = cleaner signal
const confidence = Math.min(100, snr * 8);
```

When `confidence < 60%`, a ``biosync:lowConfidence`` event fires, and the fusion model redistributes weights: rPPG drops from  $0.3 \rightarrow 0.1$ , semantic rises from  $0.2 \rightarrow 0.4$ .

The system **never fails silently** — it degrades gracefully and communicates uncertainty.

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### Frequency-Domain Analysis (Calibration Use)

**File:** `signal-processing/fftAnalysis.ts`

For calibration validation and signal quality assessment, BioSync uses the

**Goertzel algorithm** — an efficient single-frequency DFT that avoids a full FFT

when only a few target frequencies are of interest.

```
function goertzelPower(signal, targetHz, sampleRate): number {
  const omega = (2 * Math.PI * targetHz) / sampleRate;
  const coeff = 2 * Math.cos(omega);
  let s0 = 0, s1 = 0, s2 = 0;
  for (let i = 0; i < signal.length; i++) {
```

```
s0 = signal[i] + coeff * s1 - s2;
s2 = s1; s1 = s0;
}
return s1*s1 + s2*s2 - s1*s2*coeff;
}
```

Scanning from 45–150 BPM (every 1 BPM = 105 Goertzel evaluations) is faster than a full 150-point FFT for this use case. The dominant BPM provides a sanity check that the signal contains a plausible cardiac frequency — used in the confidence score.

### SNR computation:

```
SNR = power in cardiac band (0.75-2.5 Hz) / power outside cardiac band
```

SNR > 2.0 indicates good signal quality. SNR < 0.5 indicates the signal is dominated by noise — the system flags this and increases audio channel weight.

## Vocal Jitter Analysis

### *Physiological Basis*

The HPA axis pathway (above) leads to cortisol-mediated laryngeal muscle tension. This produces **cycle-to-cycle instability** in the fundamental frequency (F0) of voice.

```
Jitter (%) = [Σ|T(i+1) - T(i)| / (N-1)] / T̄ × 100
```

Where T(i) are consecutive glottal period durations (T = 1/F0, in seconds).

| Condition                        | Relative Jitter |
|----------------------------------|-----------------|
| Normal conversational speech     | 0.5–1.0%        |
| Mild stress                      | 1.0–1.5%        |
| Acute stress                     | >1.5%           |
| Extreme stress / vocal pathology | >3.0%           |

### References:

- Goberman, A.M., & Elmer, L.W. (2003). Acoustic analysis of clear versus conversational speech in individuals with Parkinson disease. *\*Journal of Communication Disorders\**, 36(2), 71–81.
- Titze, I.R. (1994). *\*Principles of Voice Production\**. Prentice Hall.

### *F0 Extraction via Autocorrelation*

**File:** hooks/useVocalJitter.ts

```
function detectPitch(buffer: Float32Array): number {
// Search lag range corresponding to F0 between 80-300 Hz
const minPeriod = floor(44100 / 300); // = 147 samples
const maxPeriod = floor(44100 / 80); // = 551 samples
let maxCorrelation = -Infinity, bestPeriod = 0;
for (let lag = minPeriod; lag <= maxPeriod; lag++) {
let r = 0;
for (let i = 0; i < buffer.length - lag; i++) r += buffer[i] * buffer[i + lag];
if (r > maxCorrelation) { maxCorrelation = r; bestPeriod = lag; }
}
```

```
// Reject unvoiced frames: normalized correlation < 0.01
const normalized = maxCorrelation / (buffer.length * 0.5);
return normalized < 0.01 ? 0 : (bestPeriod > 0 ? 44100 / bestPeriod : 0);
}
```

Voice activity detection uses RMS energy threshold ( $\text{rms} > 0.01$ ). Below threshold, jitter score decays 5 points per 500ms interval — modeling the natural return to baseline when the user is not speaking.

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## Signal Confidence and Adaptive Weight Redistribution

BioSync's fusion model adapts to signal quality degradation rather than failing.

| Condition             | Standard weights    | Adaptive weights                         |
|-----------------------|---------------------|------------------------------------------|
| All signals healthy   | J:0.5, V:0.3, I:0.2 | —                                        |
| rPPG confidence < 60% | —                   | J:0.5, V:0.1, I:0.4                      |
| No speech (J=0)       | J:0.5, V:0.3, I:0.2 | Score driven by V+I with implicit J=0    |
| Face not detected     | —                   | Confidence=0, score frozen at last value |

This ensures the system **degrades gracefully** — a weaker signal produces a less confident but still meaningful score, rather than producing garbage or crashing.

The signal confidence percentage shown in the UI communicates this honestly.

Judges who see 78% confidence understand: the system knows its own limitations.

That is a mark of engineering maturity, not weakness.